http://pubsonline.informs.org/journal/opre

OPERATIONS RESEARCH
Vol. 70, No. 3, May–June 2022, pp. 1806–1821
ISSN 0030-364X (print), ISSN 1526-5463 (online)

Methods

Always Valid Inference: Continuous Monitoring of A/B Tests

Ramesh Johari,[a] Pete Koomen,[b] Leonid Pekelis,[c] David Walsh[d]

[a] Department of Management Science and Engineering, Stanford University, Stanford, California 94305; [b] Optimizely, Inc., San Francisco, California 94105; [c] CloudTrucks, Inc., San Francisco, California 94103; [d] Unlearn.AI, San Francisco, California 94105

Contact: rjohari@stanford.edu, https://orcid.org/0000-0002-3960-0770 (RJ); pete@optimizely.com (PK); lpekelis@gmail.com, https://orcid.org/0000-0002-3126-9278 (LP); davidjmwalsh@gmail.com, https://orcid.org/0000-0003-2568-1802 (DW)

Received: December 11, 2017
Revised: May 20, 2019; July 7, 2020
Accepted: November 2, 2020
Published Online in Articles in Advance: August 10, 2021

Area of Review: Stochastic Models

https://doi.org/10.1287/opre.2021.2135

Copyright: © 2021 The Author(s)

Abstract. A/B tests are typically analyzed via frequentist *p*-values and confidence intervals, but these inferences are wholly unreliable if users endogenously choose samples sizes by *continuously monitoring* their tests. We define *always valid* *p*-values and confidence intervals that let users try to take advantage of data as fast as it becomes available, providing valid statistical inference whenever they make their decision. Always valid inference can be interpreted as a natural interface for a sequential hypothesis test, which empowers users to implement a modified test tailored to them. In particular, we show in an appropriate sense that the measures we develop trade off sample size and power efficiently, despite a lack of prior knowledge of the user's relative preference between these two goals. We also use always valid *p*-values to obtain multiple hypothesis testing control in the sequential context. Our methodology has been implemented in a large-scale commercial A/B testing platform to analyze hundreds of thousands of experiments to date.

Open Access Statement: This work is licensed under a Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International License. You are free to download this work and share with others for any purpose, except commercially, if you distribute your contributions under the same license as the original, and you must attribute this work as "*Operations Research*. Copyright © 2021 The Author(s). https://doi.org/10.1287/opre.2021.2135, used under a Creative Commons Attribution License: https://creativecommons.org/licenses/by-nc-sa/4.0/."

Funding: This research was funded by Optimizely, Inc.
Supplemental Material: The online appendix is available at https://doi.org/10.1287/opre.2021.2135.

Keywords: A/B testing • *p*-values • confidence intervals • sequential hypothesis testing • multiple hypothesis testing

1. Introduction

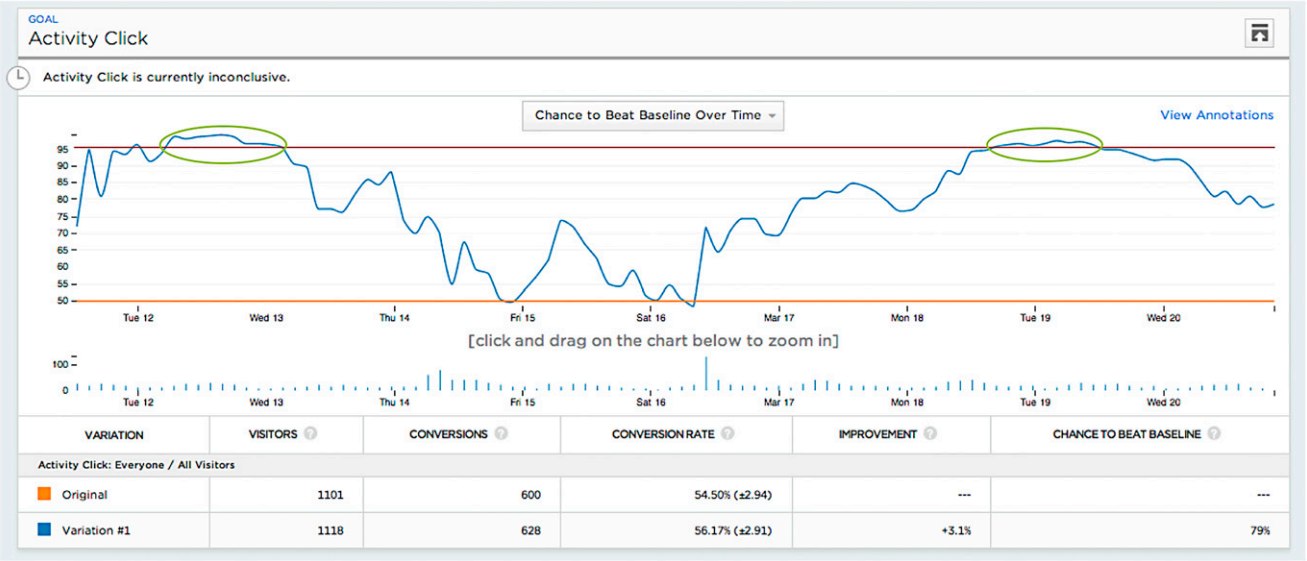
This paper reports on novel statistical methodology underlying the implementation of a large-scale commercial experimentation platform for web applications and services.[1] Web applications typically optimize their product offerings using randomized controlled trials; in industry parlance, this is known as *A/B testing*. The rapid rise of A/B testing has led to the emergence of a number of widely used platforms that handle the implementation of these experiments (Tang et al. 2010, Kohavi et al. 2013).

A standard A/B test with two variations (*control* and *treatment*) involves testing the *null hypothesis* that both groups share the same parameter (e.g., customer conversion rate from clicking on a link to a sale) against the *alternative hypothesis* that they are different. The A/B testing platform communicates experimental results to the user via frequentist parameter testing measures—that is, *p*-values and confidence intervals. Besides well-known properties of their optimality (see, e.g., Lehmann et al. 1986), these measures imply an exceptionally simple "user interface"; the user implements a mechanistic rule (threshold the *p*-value at their desired Type I error rate α) that does not require further knowledge of experiment details. This has two valuable consequences, underpinning their ubiquity in industrial practice: First, the same interface can be employed by *many* users, each with their own α; and, second, experimental results can be analyzed by users without advanced statistical training.

These desirable outcomes are only obtained, however, when *p*-values and confidence intervals are used as intended. In A/B testing practice, users are not constrained to simply analyze the output of an experiment; they can also adjust the experimental design in response to the data observed. This type of behavior can entirely undermine statistical reliability. A particularly pervasive form of this behavior is commonplace: Users *continuously monitor* the *p*-values and confidence intervals reported in order to set the sample size of an experiment dynamically (Miller 2010). Figure 1 shows how a typical dashboard enables such behavior.

The incentive to continuously monitor experiments is strong because the opportunity cost of longer experiments is large. There is value to detecting true

Figure 1. (Color online) A Typical Dashboard from a Large Commercial A/B Testing Platform Service



Notes. The graph depicts the "chance to beat baseline," measuring $1 - p_n$ over time, where p_n is the p-value, after n observations, of the null hypothesis that the clickthrough rate in treatment and control is identical. This particular test is a *A/A test*: Both the treatment and control are the *same*. The graph shows that $1 - p_n$ rises above the 95% significance threshold if the user continuously monitors the test, triggering a Type I error.

effects as quickly as possible, or giving up if it appears that no effect will be detected soon, so that they may test something else. Most users further lack good prior understanding of their cost elasticity, as well as the effect size they seek, frustrating attempts to optimize run-time in advance. Indeed, the ability to trade off maximum detection with minimum run-time dynamically is a crucial benefit of the availability of real-time data in modern A/B testing environments.

No correction for continuous monitoring is typically made in industrial practice. Consequently, the resulting feedback loop between the statistical output and the experimental design undermines inferential validity, as computations are performed under the naive assumption of their independence. Very high false-positive probabilities are obtained—well in excess of the nominal α that the user can tolerate. Even with 10,000 samples (a sample size that is quite common in online A/B testing), Type I error can easily increase fivefold.

Our challenge is the following: Can we deliver efficient inference in a simple interface, but in an environment where users continuously monitor experiments and where their priorities regarding run-time and detection are not known in advance? We develop a statistical approach that addresses this challenge and report on an implementation in a commercial A/B testing platform. A key to our framework is that we employ the *same interface* as a traditional A/B testing platform: We present the user with p-values and confidence intervals. However, our measures have the following properties. First, Type I error is controlled under any data-dependent rule the user might choose to stop the experiment (i.e., any stopping time for the data). Continuous monitoring does not inflate Type I error. Second, in a sense we make precise below, we show that if the user stops when our p-value drops below her personal α, the resulting rule approximately obtains an efficient trade-off for run-time and detection, even with no advance knowledge of her preferences.

In more detail, our contributions are as follows.

1. *Type I error control: Always valid p-values and confidence intervals*. Our first contribution is to develop p-values that control Type I error in a strong sense. In particular, we ask: What p-value processes control Type I error at any data-dependent stopping time the user might choose? We refer to such p-values as *always valid p-values* (and, analogously, *always valid confidence intervals*). We show how these always valid p-value processes can be constructed using sequential hypothesis tests (Wald 1945, Siegmund 1978, Siegmund 1985, Lai 2001); in particular, we identify a duality between always valid p-values and those sequential tests that do not accept H_0 in finite time, known as *sequential tests of power one* (Robbins and Siegmund 1974). Under this duality, the natural policy of stopping the first time that the always p-value process crosses the level α implements the corresponding sequential test of power one. In this way, we retain the simple "user interface" of p-values, but guarantee Type I error control under continuous monitoring.

2. *Efficiently trading off power and run-time: The mixture sequential probability ratio test*. Having controlled Type I error, how can we efficiently trade off power and run-time? The challenge for an A/B testing platform, as noted above, is that users' objective functions—and, in particular, their relative prioritization of run-time and detection—are not typically known in advance. We aim to find always valid p-value processes that lead to an efficient trade-off for the user.

It is evident that without restricting the class of user models we consider, no meaningful result is possible; the space of potential user preferences over run-time and detection is vast (and some are unreasonable, e.g., preferring long experiments that do not detect effects). Instead, we focus on users that generally want short tests and good detection, modeled as follows: The user stops at either the first time the p-value process crosses α or at a fixed maximum *failure time* M, whichever comes first. A larger M indicates greater patience, and a corresponding preference for detection of smaller effects. Although this class does not capture all possible objective functions, it does allow us to capture what we consider to be the most interesting axis of user heterogeneity: how much they care about power versus how much they care about run-time.

Our main contribution is to demonstrate that always valid p-values derived from a particular class of sequential tests known as *mixture sequential probability ratio tests* (mSPRTs) (Robbins 1970) achieve an efficient trade-off between power and run-time for such users to first order in M as $M \to \infty$. We achieve this result for settings where the data are generated from a single-parameter exponential family. We complement our theory with empirical analysis that compares the mSPRT to other sequential tests at finite values of M. Together, these results demonstrate that the mSPRT delivers high performance in regimes relevant for practical application.

The mSPRT involves stopping the first time a mixture of the likelihood ratio of alternative to null crosses a threshold, where the mixture is a distribution over potential values of the alternative. Although first-order efficiency as $M \to \infty$ holds for any mixing distribution, the mixing distribution plays an important role in second-order performance. In a Bayesian setting (where effect sizes are drawn from a prior), we find a particular choice of mSPRT that minimizes expected run-time in the limit where $M \to \infty$ with an appealingly simple form: When there is a Gaussian prior on effect sizes, the optimal mSPRT approximately matches the variance of the mixing distribution to the variance of the prior distribution of effect sizes. We find through numerical analysis that although the mixing distribution can have a significant impact on performance, there is also robustness, in that a well-chosen mixing distribution can deliver good performance in a wide range of conditions.

3. *Implementation in a commercial A/B testing platform*. As noted above, a key contribution of our work is the deployment of our methods in a commercial A/B testing platform, used by thousands of customers worldwide. The main technical challenge in implementation is that our results above on optimality of the mSPRT are derived in a *single-stream* setting, where a single unknown parameter of a data-generating process is being tested; by contrast, in A/B testing, because there are two variations (control and treatment), we are in a *two-stream* setting. We extend our work to this setting, essentially by treating additional parameters besides the true treatment effect as nuisance parameters; this is the method that is deployed in the platform. We also report both empirically and theoretically on a comparison of our methodology with classical fixed-horizon testing. We find that our approach can deliver equivalent power as fixed-horizon testing, with a sublinear sample size, suggesting that even statistically savvy users would receive results faster using our approach in practice.

We note further that in a companion conference paper, we provide greater detail on the implementation of our work in the large commercial A/B testing platform described above; the reader is referred to Johari et al. (2017).

4. *Multiple hypothesis testing*. Finally, we note that a major practical advantage of always valid p-values is that we can employ them within existing methodology that uses these measures as inputs to provide error guarantees in multiple hypothesis testing, despite the sequential nature of A/B testing. In particular, we can control the *family-wise error rate* (FWER), as well as the *false discovery rate* (FDR) in the sequential setting under some assumptions on the user's stopping time. We also study *false coverage rate* (FCR) control for confidence intervals. This combination of sequential testing and multiple hypothesis testing corrections is novel to our work and enabled because of the introduction of always valid p-values. The resulting FDR- and FCR-controlling procedures have also been implemented in the commercial service described above, as part of the same deployment.

Taken together, we preserve the benefits of p-values and confidence intervals, while modernizing their computation to account for continuous monitoring, as well as multiple hypothesis testing. The remainder of the paper is organized as follows. In Section 2, we present related literature. In Section 3, we describe our basic model and notation. In Section 4, we introduce always valid p-values and confidence intervals. In Section 5, we study efficient always valid p-value processes via

the use of the mSPRT, both theoretically and empirically. In Section 6, we discuss deployment and implementation, and particularly the adaptation of our basic theory to two streams of data. In Section 7, we discuss multiple hypothesis testing. Finally, we conclude in Section 8.

2. Related Work

2.1. Sequential Hypothesis Testing

The desire to test hypotheses using data that arrive sequentially over time is far from new. Rather, sequential analysis is a mature field with roots dating back to Wald (1945), and sequential tests are widely used in areas such as pharmaceutical clinical trials. For its history, methodology, and theory, we direct the reader to the encyclopedic resource of Ghosh and Sen (1991); see also Siegmund (1985) for an introduction to the topic.

In fact, there has been recent interest in implementing existing sequential tests in online experimentation (Miller 2015). However, adoption has been limited, because the tests function as "black boxes" until the single stopping time when the null hypothesis is accepted or rejected. Inference at general stopping times is not attainable. Thus, an off-the-shelf sequential test can provide value for a single experimenter only if the test is specifically optimized to her preferences over power and run-time. On the other hand, by using sequential tests as building blocks to construct always valid p-values and confidence intervals, this paper obtains a real-time interface that can better handle users with heterogeneous preferences.

2.2. Sequential Confidence Intervals and the Law of the Iterated Logarithm

Our always valid confidence intervals are not the first attempt to construct sequences of intervals that contain an unknown parameter with uniformly high probability over an infinite time horizon. Although our always valid measures emerge from a hypothesis testing framework, such intervals may be derived by appealing directly to the Law of the Iterated Logarithm (LIL), which governs the asymptotic behavior of large deviations in sample averages. This approach dates back to Darling and Robbins (1967) and Robbins (1970), where the constructed intervals are referred to as "confidence sequences." Since then, LIL-type bounds have been obtained for broader classes of (suitably normalized) martingales; for a comprehensive summary, we defer to de la Peña et al. (2008). Further, work such as Balsubramani (2014) has tightened the LIL at finite samples, while Zhao et al. (2016) extends various classical concentration inequalities to hold at data-dependent stopping times. Independent of our work in this paper, related work has leveraged these LIL-type bounds for the construction of always valid confidence intervals for a range of parametric and nonparametric settings (Balsubramani and Ramdas 2015, Howard et al. 2018).

In Section 5.6.3, we compare the empirical performance of our always valid measures derived from the mSPRT against these LIL-based intervals as vehicles for trading off power against run-time. We note that the LIL provides intervals whose size shrinks at a faster asymptotic rate than the mSPRT, and so offers better inference for users who are willing to wait until power very close to one is achieved. For most users, however, our empirical investigation suggests that the stronger finite sample performance of the mSPRT would result in a more efficient trade-off. For this reason, we consider it the most appropriate choice for an A/B testing platform.

2.3. Bandit Algorithms

Although the work described in this paper largely addresses A/B testing carried out using the toolkit of statistical inference (i.e., hypothesis testing), it is worth noting that A/B testing is often used to find the best alternative. For example, a marketing team may be interested in testing two designs for the same web page, with the only goal to maximize conversion rate. The latter type of problem has been extensively studied in the literature on *multiarmed bandits* (Bubeck et al. 2012, Lattimore and Szepesvári 2020). This has led to some use of bandit algorithms in industry in place of hypothesis testing (Scott 2015).

Two variants of the bandit-problem formulation are relevant: the *pure-exploration* problem (when the rewards during the experiment are ignored) (Bubeck et al. 2009) and the more widely studied *regret minimization* problem, where the rewards earned during experimentation are also optimized as part of the objective function. The pure-exploration problem is more relevant to the industrial experimentation scenarios that inspired this paper, as in these situations, the experimentation period is relatively short compared with the lifespan of any "winning" feature that is deployed. Therefore, the rewards earned during experimentation itself can be safely ignored. In solving the pure-exploration problem, the allocation of treatments to incoming traffic is modified dynamically, in order to provide a decision at some minimal stopping time on which treatment is the best.

We note here that much of the interest in sequential confidence intervals has arisen in connection with the literature on pure-exploration bandits, where the target error bounds are of a different flavor than the Type I error-rate control sought in this paper. Whereas we seek to control the probability of falsely detecting any treatment effect when no such effect exists, the typical focus in that literature is on *probably almost*

correct (PAC) bounds (Even-Dar et al. 2002, Kalyanakrishnan et al. 2012). There, it is only necessary to identify the winning treatment with high probability in the case that a treatment effect exists and exceeds some prespecified threshold. For instance, Kaufmann et al. (2014) use such a formulation to characterize the complexity of pure-exploration problems. Jamieson et al. (2014) and Abbasi-Yadkori et al. (2011) each present bandit algorithms, which improve regret by leveraging sequential confidence intervals that are always valid in the PAC sense. PAC-style bounds are typically not sufficient for an A/B testing platform, because the threshold on treatment effects sought cannot be tailored to the heterogeneous users.

Finally, in light of this discussion on bandits, it is worth noting why our own work emphasizes the hypothesis-testing viewpoint as opposed to a pure-exploration multiarmed bandit approach. The main point we make here is that at the same time as the best alternative is sought prospectively, there is also a postexperiment interest in *inference*; in particular, the experimenter often wants a confidence interval on the effect size of a variation that does not win. Part of the reason is operational: If the gain of a winning variation is insufficient over the status quo, the deployment cost may not be worthwhile. Such confidence intervals are also strategically important, as they provide guidance on what types of experiments may be worth trying in the future. Our observation is that the hypothesis-testing framework remains the dominant form of A/B testing in industrial deployment, at least partly for such reasons, and, for this reason, we have focused our attention on this setting. However, bandit methods are also practically valuable, and developing methods for inference with adaptive allocation remains an interesting direction for future work. We return to this point in our conclusion, Section 8.

2.4. Sequential Multiple Hypothesis Testing

There has also been recent interest in achieving multiple hypothesis testing controls in sequential contexts. For the most part, work in this area considers a different form of streaming data from the one described in this paper: Demets and Lan (1994), Foster and Stine (2008), and Javanmard and Montanari (2016) provide so-called "α-spending" and "α-investing" methods to control the family-wise error rate or the false discovery rate when experiments are performed sequentially, but within each experiment, the data are accessed only once.

However, recent work has started to address the within-experiment data-arrival process that characterizes A/B testing. Yang et al. (2017) combine α-investing with sequential hypothesis testing to enable FDR control in this regime. Jamieson and Jain (2018) go a step further, allocating traffic to treatments dynamically with the goal of achieving statistical significance quickly, while still bounding FDR. Malek et al. (2017) investigates when always valid p-values can be used to achieve other multiple testing bounds beyond FWER and FDR.

3. Preliminaries

To begin, we suppose that our data can be modeled as independent observations from an exponential family $\mathbf{X} = (X_n)_{n=1}^{\infty} \overset{iid}{\sim} F_\theta$, where the parameter θ takes values in $\Theta \subset \mathbb{R}^p$. Throughout the paper, $(\mathcal{F}_n)_{n=1}^{\infty}$ will denote the filtration generated by $(X_n)_{n=1}^{\infty}$, and $\mathbb{P}_\theta$ will denote the measure (on any space) induced under the parameter θ. Our focus is on testing a simple null hypothesis $H_0 : \theta = \theta_0$ against the composite alternative $H_1 : \theta \neq \theta_0$. (In Section 6, we adapt our analysis to two-sample hypothesis testing, as is needed to test differences between control and treatment in an A/B test.)

3.1. Decision Rules

In general, a decision rule is a pair (T, δ), where T is a (possibly infinite) stopping time for $(\mathcal{F}_n)_{n=1}^{\infty}$ that denotes the sample size at which the test is ended, and δ is a binary-valued, $(\mathcal{F}_T)$-measurable random variable, where $\delta = 1$ indicates that H_0 is rejected; note that $\delta = 0$ must hold almost surely if $T = \infty$. Note that we allow the possibility that the decision rule can be data-dependent; when T is not data-dependent, we refer to the rule as a *fixed-horizon* decision rule.

3.2. Type I Error

Type I error is the probability of erroneous rejection under the null—that is, $\mathbb{P}_{\theta_0}(\delta = 1)$. We assume that the user wants to bound Type I error at level $\alpha \in (0, 1)$.

3.3. Sequential Tests

Given α, we typically consider a family of decision rules parameterized by α. Formally, a *sequential test* is a family of decision rules $(T(\alpha), \delta(\alpha))$ indexed by $\alpha \in (0, 1)$ such that:

1. The decision rules are *nested*: $T(\alpha)$ is a.s. nonincreasing in α, and $\delta(\alpha)$ is a.s. nondecreasing in α.
2. For each α, the Type I error is bounded by α: $\mathbb{P}_{\theta_0}(\delta = 1) \leq \alpha$.

Note that sequential tests allow the possibility that the decision rules are data-dependent, though strictly speaking, fixed-horizon decision rules are allowed in this definition as well.

3.4. Fixed-Horizon Testing

Under the default fixed-horizon testing approach, we restrict to decision rules (n, δ), where the stopping time is required to be deterministic. In this setting, the objective is to maximize the power (the probability of

detection under H_1) at that n. Indeed, for data in an exponential family, for any given n, there exist a family of uniformly most powerful (UMP) tests parameterized by α, each of which maximizes power uniformly over θ among tests with Type I error rate α. These tests reject the null if a particular test statistic τ_n exceeds a threshold $k(\alpha)$ (see, e.g., chapter 4 of Lehmann et al. 1986).

While the tests maximize power for the given n, the power increases as n is increased, and so the user must choose n to trade off power against the opportunity cost of waiting for more samples. The challenge for the user is that the power is a steep function of the true θ, so good advance knowledge on the size of the effect sought is required.

3.5. The Fixed-Horizon User Interaction Model

Testing platforms typically allow users to implement their optimal test via *p-values*. Specifically, the p-value at time n corresponding to the UMP test is:

$$p_n = \inf\{\alpha : \tau_n \geq k(\alpha)\}.$$

In other words, this p-value is the smallest α such that the α-level test with sample size n rejects H_0.

The process p_n provides sufficient information for the user to implement her desired test with ease: She waits for her chosen n and rejects the null hypothesis if $p_n \leq \alpha$. In addition, p_n ensures transparency in the following sense: Because each rule $\delta_n(\alpha)$ controls Type I error at level α, any other user can threshold the p-value obtained at her own appropriate $\tilde{\alpha}$ level to satisfy her desired Type I error bound.

In fact, to control Type I error, we require only that the p-value is *superuniform*:

$$\forall s \in [0,1], \ \ \mathbb{P}_0(p_n \leq s) \leq s. \tag{1}$$

More generally, we refer to any $[0,1]$-valued, $(\mathcal{F}_n)$-measurable random variable p_n that satisfies (1) as a *fixed-horizon p-value* process for the choice of sample size n.

Confidence intervals can be constructed from the tests $\delta_n(\alpha)$ associated with fixed-horizon p-values for $H_0 : \theta = \tilde{\theta}$ at each $\tilde{\theta} \in \Theta$ by considering the set of θ that are not rejected. What matters is the following coverage bound: A $(1-\alpha)$-level fixed-horizon confidence interval is any $(\mathcal{F}_n)$-measurable random set $\mathsf{CI}_n \subset \Theta$ where

$$\forall \theta \in \Theta, \ \ \mathbb{P}_\theta(\theta \in \mathsf{CI}_n) \geq 1 - \alpha. \tag{2}$$

4. Always Valid Inference

Our goal is to let the user stop the test whenever they want, in order to trade off power with run-time as they see fit; the p-value they obtain should control Type I error. Our first contribution is the definition of *always valid* p-values as those processes that achieve this control.

Definition 1. A sequence of fixed-horizon p-values (p_n) is an always valid p-value process if given any (possibly infinite) stopping time T with respect to $(\mathcal{F}_n)$, there holds:

$$\forall s \in [0,1], \ \ \mathbb{P}_{\theta_0}(p_T \leq s) \leq s. \tag{3}$$

The following theorem demonstrates that always valid p-values are in a natural correspondence with sequential tests.

Theorem 1.

1. *Let $(T(\alpha), \delta(\alpha))$ be a sequential test. Then,*

$$p_n = \inf\{\alpha : T(\alpha) \leq n, \delta(\alpha) = 1\}$$

defines an always valid p-value process.

2. *For any always valid p-value process $(p_n)_{n=1}^\infty$, a sequential test $(\tilde{T}(\alpha), \tilde{\delta}(\alpha))$ is obtained from $(p_n)_{n=1}^\infty$ as follows:*

$$\tilde{T}(\alpha) = \inf\{n : p_n \leq \alpha\}; \tag{4}$$

$$\tilde{\delta}(\alpha) = 1\{\tilde{T}(\alpha) < \infty\}. \tag{5}$$

If $(p_n)_{n=1}^\infty$ was derived as in part (1) and $T = \infty$ whenever $\delta = 0$, then $(\tilde{T}(\alpha), \tilde{\delta}(\alpha)) = (T(\alpha), \delta(\alpha))$.

Proof of Theorem 1. For the first result, nestedness implies the following identity for any $s \in [0,1], n \geq 1, \varepsilon > 0$:

$$\{p_n \leq s\} \subset \{T(s+\varepsilon) \leq n, \delta(s+\varepsilon) = 1\} \subset \{\delta(s+\varepsilon) = 1\}.$$

Therefore:

$$\mathbb{P}_{\theta_0}(p_T \leq s) \leq \mathbb{P}_{\theta_0}(\textstyle\bigcup_n \{p_n \leq s\}) \leq \mathbb{P}_{\theta_0}(\delta(s+\varepsilon) = 1) \leq s + \varepsilon.$$

The result follows on letting $\varepsilon \to 0$. For the converse, it is immediate from the definition that the tests are nested, and $\delta(\alpha) = 0$ whenever $T(\alpha) = \infty$. For any $\varepsilon > 0$

$$\mathbb{P}_{\theta_0}(\delta(\alpha) = 1) = \mathbb{P}_{\theta_0}(T(\alpha) < \infty) \leq \mathbb{P}_{\theta_0}(p_{T(\alpha)} \leq \alpha + \varepsilon) \leq \alpha + \varepsilon,$$

where the last inequality follows from the definition of always validity. Again, the result follows on letting $\varepsilon \to 0$. □

The p-value defined in part (1) of the theorem is not the unique always valid p-value associated with that family of sequential tests (i.e., for which part (2) holds). However, among such always valid p-values, it is a.s. *minimal* at every n, resulting from the fact that it is a.s. *monotonically nonincreasing* in n. Thus, we have a one-to-one correspondence between monotone always valid p-value processes and families of sequential tests that do not give up for failure; that is, where $\delta = 0$ implies $T = \infty$. These processes can be seen as the natural representation of those sequential tests in a streaming p-value format.

4.1. The New User Interaction Model

The time $\tilde{T}(\alpha)$ represents the natural stopping time of a hypothetical user who incurs no opportunity cost from longer experiments. By thresholding the p-value at α at that time, she recovers the underlying sequential test and is able to reject H_0 whenever $\delta = 0$. Of course, a real user cannot wait forever, so she must stop the test and threshold the p-value at some potentially earlier, a.s. finite stopping time. In so doing, she sacrifices some detection power. This trade-off for the user between power and average run-time is a central concern of our proposed design and is studied in more detail in Section 5.

4.2. Confidence Intervals

Always valid confidence intervals (CIs) are defined analogously and may be constructed from always valid p-values, just as in the fixed-horizon context. Proposition 1 follows immediately.

Definition 2. A sequence of fixed-horizon $(1-\alpha)$-level confidence intervals (CI_n) is an always valid $(1-\alpha)$-level confidence interval process if, given any stopping time T with respect to $(\mathcal{F}_n)$, there holds:

$$\forall\ \theta \in \Theta,\ \ \mathbb{P}_\theta(\theta \in \mathsf{CI}_T) \geq 1-\alpha. \tag{6}$$

Proposition 1. *Suppose that, for each $\tilde{\theta} \in \Theta$, $(p_n^{\tilde{\theta}})$ is an always valid p-value process for the test of $\theta = \tilde{\theta}$. Then,*

$$\mathsf{CI}_n = \left\{\theta : p_n^\theta > \alpha\right\},$$

is an always valid $(1-\alpha)$-level CI process.

5. Efficient Always Valid *p*-Values via the mSPRT

As noted in the preceding section, users who continuously monitor experiments are making a dynamic trade-off between two objectives: detecting true effects with maximum probability and minimizing the typical length of experiments. A significant challenge for the platform is that always valid p-values must be designed without prior knowledge of the user's preferences. We are led, therefore, to consider the following design problem: How should always valid p-values be designed to lead users to an efficient trade-off between power and run-length, without access to the user's preferences in advance?

In this section, we first introduce a natural model of user behavior that encodes a trade-off between power and run-length, where users are characterized by their Type I error tolerance, α, and the maximum number of observations, M, they can afford. We assume that such a user stops and rejects H_0 at either the first time the always valid p-value crosses α or at time M, whichever comes first. We refer to such a user as a *type* (M,α) user. We study efficiency of an always valid p-value process via the *power profile* and *relative run-length profile* that the process induces for an (M,α) user across possible treatment effects; the former gives the probability that true effects are detected by this user, and the latter gives the expected run-length for such a user, normalized by M. Informally, our goal is to deliver high power at low relative run-lengths.

To achieve this goal, we consider always valid p-value processes derived from a particular family of sequential tests, the *mixture sequential probability ratio tests*. The mSPRT stops the first time a mixture (over treatment effects) of likelihood ratios against the null crosses a threshold. These were first introduced by Robbins (1970).

We begin our study in Section 5.3 by considering *first-order efficiency* of the mSPRT, in the limit where $\alpha \to 0$. For users where M is relatively large or relatively small compared with $\log(1/\alpha)$, efficiency (in an appropriate sense) is a relatively weak requirement and easily established for the mSPRT. Thus, the more interesting analysis is for users where $M \sim \log(1/\alpha)$. For such users, we establish that the mSPRT satisfies a desirable first-order efficiency property: There is no other feasible decision rule with relative run-length that is lower on some nonnull effects, a size constraint α, and power at least as high at all nonnull effects.

First-order efficiency is important, but does not tell the complete story. In particular, the preceding efficiency result does not depend on the mixing distribution employed by the mSPRT, whereas we should certainly expect that finite M (or fixed α) performance will be influenced by the choice of the mixing distribution. In Section 5.4, we theoretically investigate the importance of the choice of prior. We first theoretically investigate this question by considering a Bayesian setting where effects are drawn from a prior G. We characterize the expected run-length minimizing mixture for type (M,α) users where $M \sim \log(1/\alpha)$. Specialized to the setting of normal data, we note the intuitive result that this optimal mixture involves "matching" the prior in an appropriate sense. In Section 5.6.1, we carry out an empirical investigation of the role of the choice of mixing distribution for (M,α) users; although we find that the choice of mixture matters, we also find that the performance of the mSPRT is surprisingly robust to misspecification.

We conclude our analysis by comparing the mSPRT to other decision rules. First, in Section 5.5, we compare the mSPRT to fixed-horizon testing. In particular, we show that for (M,α) users, the mSPRT provides an improvement over fixed-horizon testing in the $\alpha \to 0$ limit, even when the fixed-horizon test is tailored to prior knowledge. In Section 5.6.2, we complement this theoretical result with empirical comparison of the mSPRT to fixed-horizon testing, demonstrating practical benefits. Second, in Section 5.6.3, we empirically

compare mSPRT performance to other always valid p-value processes derived from sequential tests (including the LIL approach discussed in Section 2) with desirable optimality criteria. These empirical results demonstrate that the mSPRT delivers high performance in the regime of M and α relevant for practical application.

5.1. The User Model

Of course, any specific user's preferences will be highly nuanced. In our analysis, for technical simplicity, we consider the following user model: We assume user preferences can be characterized by a parameter M representing the maximum number of observations that the user is willing to wait, together with their Type I error tolerance α. Given an always valid p-value process, we consider a simple model of user behavior: Users stop at either the first time the p-value drops below α (in which case they reject the null) or at time M (in which case they do not reject the null unless the p-value at time M is also below α), whichever occurs first.

We refer to such a user as a (M,α) user. In the remainder of the section, our goal will be to make near-optimal trade-offs between power and run-length for users in the limit where $\alpha \to 0$, without prior knowledge of their preferences, (M,α). Given the equivalence to a sequential test $(T(\alpha),\delta(\alpha))$, we define the (M,α) user's decision rule by $T(M,\alpha) \triangleq \min\{T(\alpha),M\}$, and remark that $\delta(M,\alpha)=1$ if and only if $T(\alpha) \le M$.

5.2. The mSPRT

The always valid p-values we employ are derived from a particular family of sequential tests: the *mixture sequential probability ratio test* (Robbins 1970). We begin by imposing slight restrictions on the data model: We assume that the data are real valued and drawn from a single-parameter exponential family, $f_\theta(x) = F'_\theta(x) = f_0(x)\exp(\theta x - \psi(\theta))$, where tests are of the natural parameter θ, Θ is an open interval, $\psi''(\theta) > 0$ for all θ, and $\mathbb{E}|X_1|^4 < \infty$. The function $\psi(\theta)$ is referred to as the *log partition function* for the family.

The mSPRT is parameterized by a *mixing distribution* H over Θ, which we restrict to have everywhere continuous and positive derivative. Given an observed sample average s_n up to time n, the likelihood ratio of θ against θ_0 is $(f_\theta(s_n)/f_{\theta_0}(s_n))^n$. Thus, we define the *mixture likelihood ratio* with respect to H as

$$\Lambda_n^H(s_n) = \int_\Theta \left(\frac{f_\theta(s_n)}{f_{\theta_0}(s_n)}\right)^n dH(\theta). \tag{7}$$

The mSPRT is then defined by:

$$T^H(\alpha) = \inf\{n : \Lambda_n^H(S_n) \ge \alpha^{-1}\}; \text{ and} \tag{8}$$

$$\delta^H(\alpha) = 1\{T^H(\alpha) < \infty\}, \tag{9}$$

where $S_n = \sum_{i=1}^n X_i/n$. The choice of threshold α^{-1} on the likelihood ratio ensures that Type I error is controlled at level α, via standard martingale techniques (Siegmund 1985). Intuitively, $\Lambda_n^H(S_n)$ represents the evidence against H_0 in favor of a mixture of alternative hypotheses, based on the first n observations. The test rejects H_0 if the accumulated evidence ever becomes large enough.

Our first motivation for considering mSPRTs is that they are *tests of power one* (Robbins and Siegmund 1974): For all α and $\theta \ne \theta_0$, there holds:

$$\mathbb{P}_\theta(T(\alpha) < \infty, \delta(\alpha) = 1) = 1.$$

In other words, for the hypothetical user who can wait forever, any mSPRT delivers power one for any alternative. Second, mSPRTs have been studied from a decision-theoretic framework, where the cost of longer experiments is taken to be the terminal sample size multiplied by some cost c per observation. The goal there is to balance this penalty against the costs associated with Type I and Type II errors. mSPRTs are found to be asymptotically optimal as $c \to 0$ (Lai 2001). This is a large data limit, analogous to the $\alpha \to 0$, $M \to \infty$ setup of this paper.

For later reference, we note the following result due to Pollak and Siegmund (1975): For any mixing distribution H, as $\alpha \to 0$,

$$T^H(\alpha)/\log(1/\alpha) \to I(\theta,\theta_0)^{-1} := \{(\theta-\theta_0)\psi'(\theta) - (\psi(\theta) - \psi(\theta_0))\}^{-1}, \tag{10}$$

holds in probability and in $\mathcal{L}^2$, where $\psi(\theta)$ is the log-partition function for the family F_θ. This result characterizes the run-length of the mSPRT in the small α limit and plays a key role in our subsequent study of efficiency.

5.3. First-Order Efficiency in the $\alpha \to 0$ Limit

We now formalize our study of the power and run-length trade-off for an (M,α) user. The set of $(T(M,\alpha),\delta(M,\alpha))$ decision rules is referred to as the set of *feasible* decision rules for an (M,α) user.

We first map the two objectives of the user to formal quantities of interest. First, an (M,α) user will want to choose her decision rule to maximize the *power profile* $\nu(\theta) = \mathbb{P}_\theta(\delta = 1)$ over $\theta \ne \theta_0$. Second, she will want to minimize the *relative run-length profile*—that is, the run-length measured against the maximum available to her, $\rho(\theta) = \mathbb{E}_\theta(T)/M$, viewed as a function of θ.

Perfect efficiency would entail $\rho(\theta) = 0$ and $\nu(\theta) = 1$ for all $\theta \ne \theta_0$. Of course, perfect efficiency is generally unattainable for feasible decision rules. In this section, we study the best achievable performance a user can hope for, in the limit where $\alpha \to 0$.

Our analysis depends on the characterization of run-length of the mSPRT in (10). The consequence is that, in the limit as $\alpha \to 0$, the study of efficiency divides into three cases, depending on the relative values of M and $\log(1/\alpha)$. We consider these cases in turn.

5.3.1. "Aggressive" Users. $M \gg \log(1/\alpha)$. Users in this regime are aggressive; α is large relative to the maximum run-length they have chosen. In this regime, any mSPRT asymptotically recovers perfect efficiency in the limit where α is small. Intuitively, because the user is willing to wait a substantially longer time than $\log(1/\alpha)$, a sublinear fraction of their maximum run-length is used by the mSPRT; because the mSPRT is a test of power one, this means the user receives power near 100% in return for a near-zero run-length profile. The proof follows immediately from (10).

Proposition 2. *Given any mixture H, let $\rho(\theta)$ and $\nu(\theta)$ be the relative run-length and power profiles, respectively, associated with $(T^H(M,\alpha), \delta^H(M,\alpha))$. If $\alpha \to 0$ and $M \to \infty$ such that $M/\log(1/\alpha) \to \infty$, we have $\rho(\theta) \to 0$ and $\nu(\theta) \to 1$ at each $\theta \neq \theta_0$.*

"Conservative" Users: $M \ll \log(1/\alpha)$. Users in this regime are conservative; α is small relative to the maximum run-length they have chosen. In this case, experimentation is not productive: The user is unwilling to wait long enough to detect any effects. Thus, any mSPRT trivially performs as well as any feasible decision rule for the user.

Proposition 3. *For each (M,α), and any feasible decision rule, let ν be the associated power profile. Then, if $\alpha \to 0, M \to \infty$ such that $M/\log(1/\alpha) \to 0$, we have $\nu(\theta) \to 0$ for each θ.*

"Goldilocks" Users: $M \sim \log(1/\alpha)$. This is the interesting case, where experimentation is worthwhile, but statistical analysis is nontrivial. To proceed, we require an additional definition. For a family of sequential tests, we want to define a measure of the worst-case efficiency over $\theta \neq \theta_0$ for an (M,α) user. Informally, we define this as the relative efficiency of the truncated test they obtain by minimizing the relative run-length everywhere, compared with any other test that offers at least as good power everywhere. This is formalized as follows.

Definition 3. Given a sequential test $(T(\alpha), \delta(\alpha))$, let $\rho(\theta;\alpha,M)$ and $\nu(\theta;\alpha,M)$ be the relative run-length and power profiles associated with $(T(M,\alpha), \delta(M,\alpha))$. The relative efficiency of this test at (M,α) is

$$\phi(M,\alpha) = \inf_{(T,\delta)} \inf_{\theta \neq \theta_0} \frac{\rho(\theta)}{\rho(\theta;\alpha,M)},$$

where the infimum is taken over all weakly more powerful, feasible decision rules; that is, over tests such that $T \leq M$ a.s., $\mathbb{P}_{\theta_0}(\delta = 1) \leq \alpha$, and for all $\theta \neq \theta_0$ there holds $\nu(\theta) \geq \nu(\theta;\alpha,M)$.

Our main result demonstrates that in the regime where $M \sim \log(1/\alpha)$, any mSPRT has relative efficiency approaching unity when $\alpha \to 0$.

Theorem 2. *Suppose ψ'' is absolutely continuous, and there is an open interval around θ_0, where $\psi''(\theta) < \infty$. Given any H, let $\phi(M,\alpha)$ be the efficiency of the mSPRT $(T^H(\alpha), \delta^H(\alpha))$. If $\alpha \to 0, M \to \infty$ such that $M = O(\log(\alpha^{-1}))$, we have $\phi(M,\alpha) \to 1$.*

Note that this result is not dependent on the mixing distribution H. This is a consequence of the fact that we study efficiency only to first order in the limit as $\alpha \to 0$. However, as we find in the next section, the choice of prior can have an important second-order effect on performance. (Note that we expect similar efficiency results to hold for the LIL-type bounds discussed in Section 2 (Balsubramani and Ramdas 2015, Howard et al. 2018).)

5.4. The Role of the Mixing Distribution *H*

In this section, we investigate the impact of the choice of the mixing distribution H on performance. As noted in the preceding section, it is users with $M \sim \log(1/\alpha)$ where there is a meaningful trade-off between run-length and power; therefore, we focus our attention on the role of H for this class of users. Theorem 2 establishes that no other test can offer a uniform improvement at every θ to first order in that limit. However, the choice of H does influence the trade-off between performance at different effect sizes, and this trade-off will yield a relevant second-order effect on performance.

In a Bayesian setting, where we have a prior $\theta \sim G$ under H_1, Theorem 3 gives the mixing distribution H that minimizes the relative run-length on average over this prior for a Goldilocks user with parameters (M,α) (i.e., a user with $M \sim \log(1/\alpha)$). This theorem provides a general characterization of one notion of an "optimal" mixing distribution; we leverage the theorem subsequently to study the optimal mixture in the case of normal data.

Theorem 3. *Suppose G is absolutely continuous with respect to the Lebesgue measure, and let H_γ be a parametric family, $\gamma \in \Gamma$, with density h_γ positive and continuous on Θ. Let $\rho_\gamma(\theta)$ be the profile of relative run-lengths associated with $(T^{H_\gamma}(M,\alpha), \delta^{H_\gamma}(M,\alpha))$. Then up to $o(1)$ terms as $\alpha \to 0$ and with $M = O(\log(1/\alpha))$, the average relative run-length $\mathbb{E}_{\theta\sim G}\{\rho_\gamma(\theta)\}$ is minimized by any γ^* such that*

$$\gamma^* \in \arg\max_{\gamma\in\Gamma} \mathbb{E}_{\theta\sim G} \mathbf{1}_{A(M,\alpha)} I(\theta,\theta_0)^{-1} \log h_\gamma(\theta), \tag{11}$$

where $A(M,\alpha) = \{\theta : I(\theta,\theta_0) \geq \log(1/\alpha)/M\}$.

If $h_\gamma(\theta) = q(\gamma_1, \gamma_2)e^{\gamma_1\theta - \gamma_2\psi(\theta)}$ is a conjugate prior for f_θ, the data distribution, then γ^* solves:

$$\frac{\partial q(\gamma_1,\gamma_2)/\partial\gamma_1}{\partial q(\gamma_1,\gamma_2)/\partial\gamma_2} = \frac{\mathbb{E}_{\theta\sim G}\mathbf{1}_A\theta I(\theta,\theta_0)^{-1}}{\mathbb{E}_{\theta\sim G}\mathbf{1}_A\psi(\theta)I(\theta,\theta_0)^{-1}}.$$

Consistent with our finding of first-order optimality for any choice of H in Theorem 2, it follows from our proof that the choice of γ does not impact $\mathbb{E}_{\theta\sim G}\{\rho_\gamma(\theta)\}$ to first order.

Heuristically, the mSPRT rejects H_0 when there is sufficient evidence in favor of any $H_1 : \theta \neq \theta_0$, weighted by the distribution H over alternatives. Thus, we might expect optimal sampling efficiency when H is *matched* to the prior. If we specialize to normal data, $f_\theta(x) = \phi(x-\theta)$, a centered normal prior, $G(\theta) = \Phi(\theta/\tau)$, and consider normal mixtures, $h_\gamma(\theta) = (1/\gamma)\,\phi(\theta/\gamma)$, this intuition is mostly accurate. (Here, we use ϕ to denote the normal probability distribution function and Φ to denote the normal cumulative distribution function.) In that case, the optimal choice of mixing variance becomes:

$$\gamma^{2*} = \tau^2\frac{\Phi(-b)}{\frac{1}{b}\phi(b) - \Phi(-b)}, \tag{12}$$

for $b = \left(2\log\alpha^{-1}/M\tau^2\right)^{1/2}$, equal to the prior variance multiplied by a factor correcting for anticipated truncation. (It is straightforward to check that γ^* is positive by noting the following properties of the denominator: It approaches infinity as $b \to 0$, has negative derivative for all b, and approaches zero as $b \to \infty$.) Sampling efficiency is improved by weighting toward larger effects when few samples are available and smaller effects where there are ample data.

We subsequently investigate the importance of the choice of H empirically in Section 5.6.1.

5.5. Comparison with Fixed-Horizon Testing

We now theoretically compare decisions based on mSPRT p-values with fixed-horizon testing. In general, the fixed-horizon sample size must be chosen in reference to the effect sizes where detection is needed; therefore, we compare performance of the mSPRT to a fixed-horizon test that is calibrated to have good average power over the prior G.

For convenience, we focus on the normal case $G = N(0,\tau^2)$, and fix $H = N(0,(\gamma^*)^2)$. We consider two rival tests for an arbitrary Goldilocks user: the mSPRT truncated at M and the fixed-horizon test chosen to have the same average power over this prior. For small α, the former has lower average relative run-length over G, as formalized in the following proposition.

Proposition 4. *Suppose $G = N(0,\tau^2)$ and $H = N(0,(\gamma^*)^2)$ (cf. Theorem 3), and let $v^*(\theta;\alpha,M)$ and $\rho^*(\theta;\alpha,M)$ be the power profile and relative run-length profile of the resulting mSPRT truncated at M. Let n^* be the sample size such that the UMP fixed-horizon test at sample size n^* has expected power matching the truncated mSPRT—that is, equal to $\mathbb{E}_{\theta\sim G}[v^*(\theta;\alpha,M)]$. Let $\rho_f(\theta;\alpha,M)$ be the relative run-length profile of this fixed-horizon test.*

If $\alpha \to 0, M \to \infty$, such that $M = O(\log(1/\alpha))$, then $\mathbb{E}_{\theta\sim G}[\rho^*(\theta;M,\alpha)]/\mathbb{E}_{\theta\sim G}[\rho_f(\theta;\alpha,M)] \to 0$.

We empirically compare the mSPRT decision rule to fixed-horizon testing with single-stream data in Section 5.6.2. In Section 6.2, we make this comparison using experiment data from the large-scale commercial A/B testing platform, where these methods were deployed.

5.6. Empirical Analysis

In this subsection, we complement our preceding theoretical analysis with a number of empirical analyses. For all our simulations, we assume the setup leading to (12)—that is, normal data with $f_\theta(x) = \phi(x-\theta)$, a centered normal prior with $G(\theta) = \Phi(\theta/\tau)$, and normal mixing distribution with $h_\gamma(\theta) = (1/\gamma)\,\phi(\theta/\gamma)$. Informally, our simulations evaluate performance of different decision rules for (M,α) users, on average, over effect sizes drawn from the prior. Formally, for each tuple (α, M, τ), we draw effect sizes $\theta_b \sim G$, $b = 1,\ldots,B$, and we simulate a stream of data $\mathbf{X}_b = (X_n)_{n=1}^M \overset{iid}{\sim} F_{\theta_b}$. We estimate average power and run-length profiles by $\hat{v} = \frac{1}{B}\sum_{b=1}^B \mathbf{1}_{\{T_b\le M\}}$ and $\hat{\rho} = BM^{-1}\sum_{b=1}^B \min\{T_b, M\}$, where T_b corresponds to the first rejection time of test $T(\alpha)$ for $H_0 : \theta = 0$ with data $\mathbf{X}_b$.

We select (α, M, τ) over the grid generated by the outer product of $\alpha \in \{10^{-4}, 10^{-2}, 10^{-1}\}$, $\tau^2 \in \{10^{-4}, 10^{-2}, 10^{-1}\}$, $M \in \{10^1,\ldots,10^7\}$, and $B = 10^4$. Rough estimates suggest these are enough Monte Carlo simulations for low variability. The effect-size distributions were designed to approximate reality for online experimenters, as $\tau^2 = 10^{-4}$ roughly matches a 10% relative improvement on a 1% conversion rate.

To achieve comparison across the wide range of parameters, we present results relative to the reparameterized maximum run-length

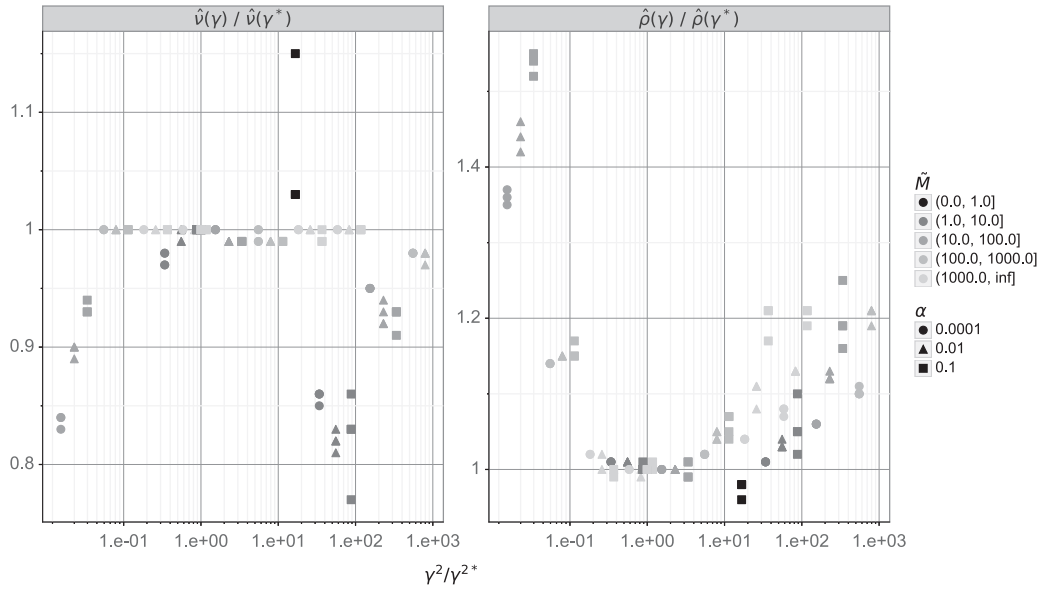
$$\tilde{M} = M\left(\frac{\log\alpha^{-1}}{\tau^2}\right)^{-1}.$$

5.6.1. The Role of the Mixing Distribution *H*: Empirics.

Nominally, the dependence of γ^* on M presents a challenge. However, as we now demonstrate, the choice of mixing distribution is quite robust with respect to variation in M, though getting γ "in the ballpark" is important.

We simulate the mSPRT for five different mixing regimes, $\gamma \in \{\gamma^*, 1, 10^{-1}\tau, \tau, 10\tau\}$. Figure 2 depicts results. Misspecification in γ up to one order of magnitude leads to a less than 5% drop in average power and a 10% increase in average run-length. Missing γ^* by two

Figure 2. Investigating the Role of Mixing Distribution H for the mSPRT



Notes. Four alternative mixing regimes are compared, $\gamma \in \{1, 10^{-1}\tau, \tau, 10\tau\}$, to γ^* (cf. Theorem 3), showing robustness in relative power ($\nu(\gamma)/\nu(\gamma^*)$) and run-length ($\rho(\gamma)/\rho(\gamma^*)$) to mixing misspecification. Note: We do not show results for severely underpowered cases ($\hat{\nu} < 0.1$), as the variance in estimates distracts from the overall picture.

orders of magnitude can result in a 20% drop in average power and a 40% increase in average run-length. Note that we do not show results for severely underpowered parameter combinations ($\hat{\nu} < 0.1$), as the variance in estimates distracts from the overall picture.

The story does change for "non-Goldilocks" users. As $\tilde{M}$ grows, $\hat{\nu} \to 1$ regardless of γ, resulting in muted gains from H optimization, while $\hat{\rho}$ shows little sensitivity to $\tilde{M}$. Lastly, we remark on the few cases where $\gamma \neq \gamma^*$ achieves superior results. These all have $\alpha = 0.1$ and $\tilde{M} < 1.0$, indicating a horizon for the breakdown of asymptotics used in Theorem 3 for finite sample regimes.

5.6.2. Empirical Comparison with Fixed-Horizon Testing. Recall that Proposition 4 gives the improvement in expected run-length of the mSPRT decision rule over an optimized fixed-horizon test, asymptotically as $\alpha \to 0$. We now evaluate this result in a finite-sample setting. Figure 3 shows that the benefit from stopping early outweighs the cost of additional slack in always valid decision boundaries at reasonable power levels ($\hat{\rho} \geq 0.5$).

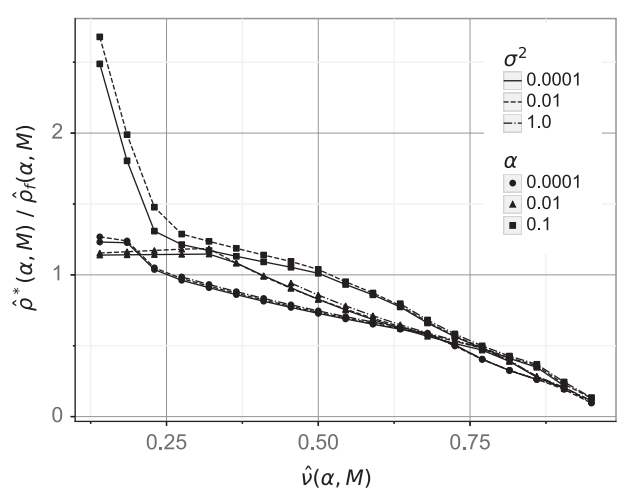
We also make a similar comparison in Section 6.2, using data from more than 10,000 experiments with two streams (treatment and control) from the large-scale commercial A/B testing platform where these methods were deployed.

5.6.3. Empirical Comparison with Other Sequential Testing Approaches. As noted above, asymptotic first-order efficiency is almost certainly not unique to the mSPRT, even across (M, α) users. The discussion in Section 2.2 of our paper, and, in particular, section 4.1.1 of Kaufmann et al. (2014), highlights tests of the form:

$$T^{\beta}(\alpha) = \inf\left\{n : S_n > \left(\frac{2\beta(n,\alpha)}{n}\right)^{1/2}\right\},$$

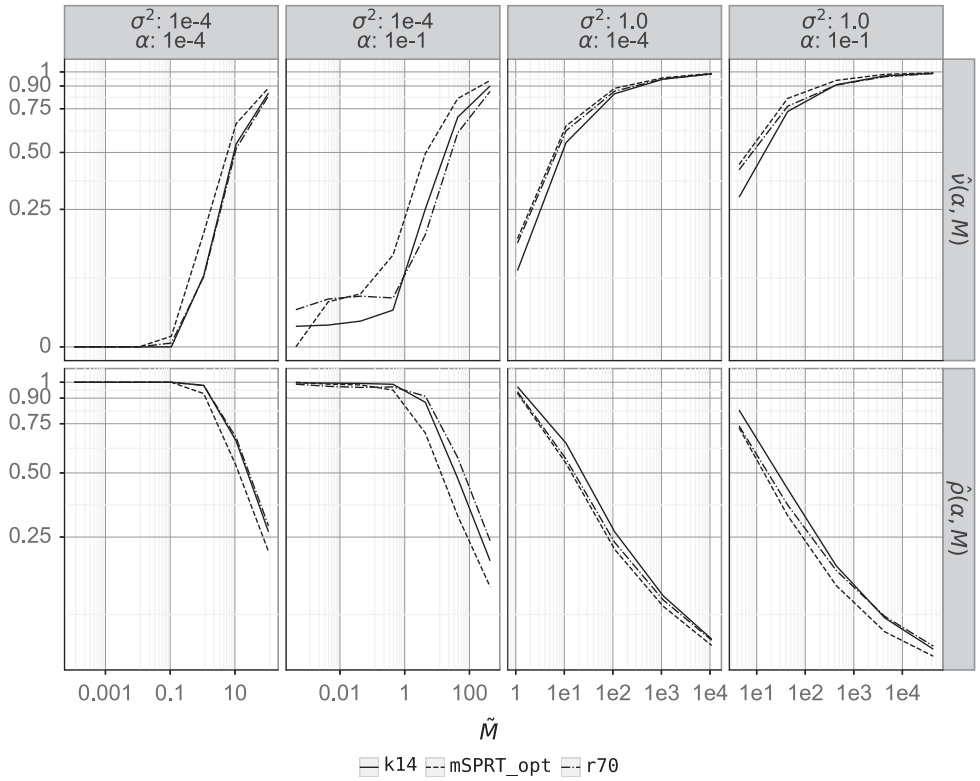
as a reasonable alternative class of candidates. We compare the mSPRT to two tests of the preceding form, from Robbins (1970) and Kaufmann et al. (2014),

Figure 3. Average Run-Time for Optimally Tuned mSPRT (ρ^*) and Fixed-Horizon UMP Test (ρ_f), Chosen to Have Same Average Power over Distribution of Effect Sizes



Note. The asymptotic result of Proposition 4 is shown to hold in many finite sample settings.

Figure 4. Average Power (ν) and Relative Run-Length (ρ) over a Distribution of Effect Sizes, Compared for Three Always Valid Decision Boundaries: The H-Optimal mSPRT (`mSPRT_opt`), a Proposal in Robbins (1970) (`r70`), and an LIL-Based Test from Kaufmann et al. (2014) (`k14`)



Note. The optimally tuned mSPRT is strictly more efficient at all but the smallest levels of truncation and continues to be substantially so even at very large truncation, despite the improved rate of efficiency gain for `k14`.

respectively. Formally, within the simulation framework specified above, we compare decision rules for (M, α) users derived from the following sequential tests:

1. The H-optimal mSPRT derived in this paper (denoted `mSPRT_opt` in the plots);
2. The test proposed in section 3 of Robbins (1970), characterized by $\beta(n, \alpha) = n + 1/n\log(n + 1/2\alpha)$ (denoted `r70` in the plots); and
3. An "LIL-based" test with $\beta(n, \alpha) = \log(\alpha^{-1}) + 3\log\log(\alpha^{-1}) + 3/2\log\log(e * n)$ from Kaufmann et al. (2014) (denoted `k14` in the plots).

Figure 4 shows that `mSPRT_opt` has optimal average power and run-length, while `r70` and `k14` vie for dominance across all finite-sample regimes examined. These results highlight a general phenomenon. Although multiple decision rules may make the Goldilocks user perfectly efficient in the limit, they differ in preferential treatment for some (M, α) users over others. The improved *rate* of efficiency gain for `k14` comes at the cost of lower efficiency for low to moderate $\tilde{M}$.

The mSPRT is given an advantage, as it is optimally tuned to the user's M via γ^*, but we stress that, although some tuning is advised, finite-sample efficiency is practically robust to H misspecification. Even for users who truncate at 100× the typical run-length ($\tilde{M} = 100$), `mSPRT_opt` has roughly 5% improvement, on average, power over its peers, which is within the two-orders-of-magnitude misspecification range identified in Section 5.6.1. Average run-length remains 20% better for all $\tilde{M} \geq 10$, and up 40% better in some cases ($\tau = 10^{-2}, \alpha = 0.1$).

6. Deployment

One technical challenge remains before the always valid p-values and confidence intervals derived in the previous section may be used in A/B testing. Although those measures address inference for a single parameter in a single independent and identically distributed (IID) sequence of data, incoming visitors to an A/B test are actually randomized into two streams, where each stream receives a distinct treatment. In this section, we summarize how the measures can be modified to address the two most typical goals in practice: inference on the difference in means between two streams of normally distributed data, and inference on the difference in success probabilities for binary-valued data (both of these are commonly referred to as the difference in "conversion rates"). The discussion is

limited to p-values, but the results may be extended to confidence intervals in the usual way. Since January 2015, these two-stream p-values and confidence intervals have been implemented in a large-scale platform serving thousands of clients, ranging from small businesses to large enterprises.

In the case of normal data, we develop a two-stream mSPRT, which gives exact uniform Type I error control for testing the composite null hypothesis that the two means are equal. Extending the asymptotic theory of the previous section, we find that first-order efficiency for trading off power and run-time in the $\alpha \to 0$ limit is still obtained. As in the single-stream case, our two-stream mSPRT is parameterized by a mixing parameter H over the unknown treatment effect. Theorem 3 carries over to this setting, indicating how this mixture should be tailored to the distribution of anticipated effects, in order to obtain good performance at moderate α. Much of the technical details are deferred to Online Appendix B.

For binary data, our two-stream mSPRT achieves approximate uniform Type I error control by appealing to the Central Limit Theorem. In this case, we use empirical data from our deployment to detail the efficiency gain over fixed-horizon testing. As in Sections 5.5 and 5.6.2, we see that the mSPRT, which is optimized to a prior for the treatment effect, can trade off power and run-time better than a comparably optimized fixed-horizon test. Of particular practical importance, the mSPRT is seen to outperform any fixed-horizon test that the experimenter might select herself, unless she has far better prior information than the platform does.

6.1. Two-Stream *p*-Values

We represent observations in the two streams by IID sequences $\mathbf{X} = (X_n)_{n=1}^{\infty}$ and $\mathbf{Y} = (Y_n)_{n=1}^{\infty}$. For normal data, we have $X_n \sim N(\mu_0, \sigma^2), Y_n \sim N(\mu_1, \sigma^2)$, where μ_0 and μ_1 are unknown, while the variance σ^2 is assumed known and common to both streams. For binary data, $X_n \sim Bernoulli(p_0), Y_n \sim Bernoulli(p_1)$, where $p_0, p_1 \in (0,1)$ are unknown. The conversion-rate difference is $\theta = \mu_1 - \mu_0$ or $\theta = p_1 - p_0$, respectively. We want to test the composite null hypothesis $H_0 : \theta = 0$ against $H_1 : \theta \neq 0$.

We make the simplification that visitors arrive in pairs with one visitor assigned to each treatment, so that observations are obtained as a sequence of pairs $(W_n)_{n=1}^{\infty} = (X_1, Y_1), (X_2, Y_2), \ldots$. An always valid p-value is understood as a process adapted to the filtration generated by this sequence, which controls Type I error uniformly over both the composite null hypothesis and the choice of stopping time. This model closely approximates the treatment allocation typically adopted in practice, where visitors arrive individually and each visitor is allocated to each treatment with 50% probability independently of all other visitors. A similar approach can be used even if the allocation to each group is not even, as long as it is fixed in advance. Extensions to other allocation policies, such as data-dependent bandit schemes, will be the subject of future work.

For normal data, we view $(W_n)_{n=1}^{\infty}$ as a single stream of IID data from a bivariate distribution parameterized by the pair (θ, μ), where $\mu = (\mu_0 + \mu_1)/2$. It is straightforward to show that, after fixing $\mu = \mu^*$ arbitrarily, that this distribution corresponds to the one-parameter exponential family $f_\theta(w) \propto \phi\left(y - x - \theta/\sigma\sqrt{2}\right)$, where $w = (x, y)$. Hence, we may implement the mSPRT based on f_θ—that is, we threshold the mixture likelihood ratio given in (7) with $s_n = 1/n \sum_{i=1}^{n} w_i$. For any μ^*, this mSPRT controls Type I error for testing the simple null $H_0 : \theta = 0, \mu = \mu^*$ against $H_1 : \theta \neq 0, \mu = \mu^*$, and so the p-value derived from this mSPRT is always valid for testing the composite null hypothesis. In Online Appendix B, we show that it satisfies natural analogues of the single-stream optimality results described in Section 5.

Unfortunately for binary data, the distribution of W_n does not reduce to a one-parameter exponential family. Nonetheless, we set $p = (p_0 + p_1)/2$ and denote the density of W_n by $f_{\theta,p}$. Then, for any θ and p^*, in the limit as $n \to \infty$, the likelihood ratio against the pair (θ_0, p^*) in favor of (θ, p^*) approaches $(\tilde{f}_\theta(s_n)/\tilde{f}_{\theta_0}(s_n))^n$, where (with $w = (x, y)$):

$$\tilde{f}_\theta(w) = \phi\left(\frac{y - x - \theta}{\sqrt{p_0^*(1-p_0^*) + p_1^*(1-p_1^*)}}\right), \quad p_0^* = p^* - \theta/2,$$
$$p_1^* = p^* + \theta/2.$$

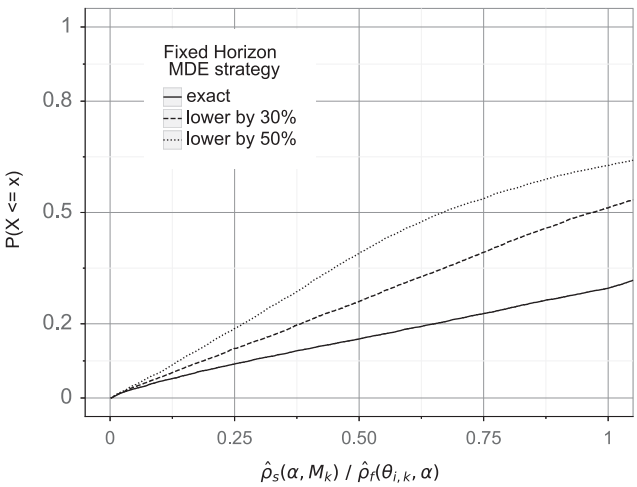
We compute the mSPRT p-values based on this density using the sample means in each stream as plug-in estimates for p_0^* and p_1^*. If α is moderate, the mSPRT terminates with high probability before this asymptotic distribution becomes accurate, so Type I error is not controlled. However, for small α, simulation shows that these p-values are approximately always valid.

6.2. Real-World Improvement

Now, we use empirical data to document the improvement of our two-stream mSPRT over fixed-horizon testing for binary data. For this purpose, we randomly sampled from binary experiments run on a large-scale commercial A/B testing platform in early 2015.

We found that the observed data were consistent with a normal distribution of true effect sizes $\theta/\sqrt{p_0^*(1-p_0^*) + p_1^*(1-p_1^*)}$ across experiments. To estimate the true effect sizes, we shrunk the distribution of observed effect sizes via James–Stein estimation

Figure 5. The Empirical Distribution of Sample Size Ratios Between the mSPRT and Various Fixed-Horizon Testing Strategies over 15,000 Real-World Experiments



Note. Even when experiments are given a priori knowledge of their effect size up to 30% error, the H-suboptimal mSPRT achieves run-time improvement a majority of the time.

(James and Stein 1961) to address the statistical noise in these observed values, $\hat{\theta}_{JS}$.

After removing all experiments with fewer than 10,000 observations, and those that were stopped before a fixed-horizon test achieved 80% power at $\hat{\theta}_{JS}$, we were left with 15,246 moderately long-running, high-powered tests. Across these, the median number of observations was 42,000 and effect size 0.01%.

In Figure 5, we compare the run-time of the mSPRT, $\rho_s(\alpha, M)$, using the two-stream approximation of the previous section and a suboptimal normal mixing distribution, H, to fixed-horizon testing strategies that the experimenter might choose if she has additional information about the effect size sought. We suppose that she can a priori estimate the unknown effect up to some specified relative error, ε, and then she selects the sample size that provides 80% power at her lower bound for the effect, $\rho_f(\alpha, \hat{\theta}_J S, \varepsilon) = \min\{n : v(\alpha, \hat{\theta}_{JS}(1-\varepsilon), n) >= 0.8\}$. For each strategy, we further removed experiments that were underpowered, according to the experimenter's fixed-horizon sample size ρ_f, and set M to achieve 80% average power.

The mSPRT achieves a run-time improvement on a majority of experiments, unless the experimenter has a very precise estimate of her effect size. In fact, a relative error below 50%, within 0.005% effect size according to our sample, would rarely be achievable in practice. Because experiment throughput is of primary importance to digital experimenters, having more than 50% of experiments terminate sooner and with a significant result is very beneficial. Further discussion, and broader practical gains associated with our two-stream mSPRT p-values, is given in our companion paper, Johari et al. (2017).

7. Multiple Testing

Always valid p-values and confidence intervals may be combined with existing multiple testing procedures when several experiments are conducted simultaneously. One option is to derive inference measures for each test individually, and these bound the expected proportion of the experiments that incur a Type I error—that is, the false-positive rate. However, that approach can be insufficient if the combination of Type I errors across multiple experiments can have a disproportionate impact on the user's ability to make good decisions. In the multiple-testing literature, fixed-horizon p-values and confidence intervals are taken as input, and the procedures output *q-values* and corrected confidence intervals that are designed to satisfy a global error constraint that better reflects the overall cost to the user.

Obtaining the same error controls with a data-dependent sample size is highly nontrivial, even if the stopping rule is fixed by the platform. However, always validity provides an opportunity to do so, while still offering the user substantial latitude to choose her own stopping time. We provide a comprehensive analysis of multiple testing within the context of always valid inference. We consider two leading multiple-testing error constraints: family-wise error rate and false discovery rate. We obtain conditions on the user's stopping time that ensure these objectives can be bounded by supplying always valid p-values and confidence intervals as input to fixed-horizon procedures in popular use. In this sense, we say that these procedures, as well as the error constraints, *"commute" with always validity* over a class of stopping times. The resulting always valid *q-values* and corrected confidence intervals have both been adopted in the large-scale commercial A/B testing platform, in appropriate contexts. Because of space constraints, full discussion of these results is provided in Online Appendix C.

8. Conclusion

Our paper derives always valid p-values and confidence intervals for A/B testing. These allow heterogeneous users to continuously monitor their experiments and to derive inferences at any time, which efficiently trade off power and run-time for their needs. In addition, we have identified how these measures may be combined with multiple hypothesis testing corrections to achieve sequential multiple testing controls.

We have only needed to place weak assumptions on the data-generating processes—namely, that observations are binary or normally distributed and independent across successive visitors entering the

experiment. However, in some contexts, A/B testing data can be heavily right-tailed (Fithian and Wager 2014). Various methods for modeling these skewed data are used in practice, and construction of always valid inference measures under these models would be a useful extension to this paper. Further, the assumption of independence can fail due to seasonal effects, which induces correlations between visitors who arrive at similar times during the experiment. Our implementation paper (Johari et al. 2017) provides a simple heuristic (a "reset policy") that identifies when seasonality may have led to inaccurate inference measures and applies conservative corrections to address this. In future work, we aim to tackle the issue in more detail, deriving inference measures from an extension of the mSPRT that models the time dependence directly.

This paper has only considered the case where incoming visitors are randomized to treatments independently of the data collected so far. In fact, it can be shown that the inference measures achieve Type I error control at any stopping time, even if treatments are assigned according to a bandit algorithm (see Section 2). Unifying these measures with bandit approaches would be valuable future work. In particular, one might ask what allocation policy enables the mSPRT to achieve significance most quickly, or how the power and relative run-time profiles of always valid measures are impacted if the allocation is optimized to another objective, such as regret minimization.

Acknowledgments

The authors thank the team at Optimizely, and in particular, Darwish Gani, Ajith Mascarenhas, Sonesh Surana, and Hao Xia, who were instrumental in the deployment of the ideas in this work. Ramesh Johari was a technical advisor to Optimizely, Inc., when the initial version of this work was carried out. Pete Koomen is co-founder at Optimizely, Inc. Leonid Pekelis and David Walsh were employed by Optimizely, Inc., when the initial version of this work was carried out.

Endnote

[1] The platform is named *Optimizely* (http://www.optimizely.com). The methodology of this paper forms the core of the statistical backend of the Optimizely platform. Throughout the paper, we refer to the platform as "a large commercial A/B testing platform" or similar.

References

Abbasi-Yadkori Y, Pál D, Szepesvári C (2011) Improved algorithms for linear stochastic bandits. *Adv. Neural Inform. Processing Systems* 24:2312–2320.

Balsubramani A (2014) Sharp finite-time iterated-logarithm martingale concentration. Preprint, submitted May 12, https://arxiv.org/abs/1405.2639.

Balsubramani A, Ramdas A (2015) Sequential nonparametric testing with the law of the iterated logarithm. Preprint, submitted June 10, https://arxiv.org/abs/1506.03486.

Bubeck S, Cesa-Bianchi N (2012) Regret analysis of stochastic and nonstochastic multi-armed bandit problems. *Foundations Trends Machine Learn.* 5(1):1–122.

Bubeck S, Munos R, Stoltz G (2009) Pure exploration in multi-armed bandits problems. Gavaldà R, Lugosi G, Zeugmann T, Zilles S, eds. *Internat. Conf. Algorithmic Learning Theory*, *Lecture Notes in Computer Science*, vol. 5809 (Springer, Berlin), 23–37.

Darling D, Robbins H (1967) Confidence sequences for mean, variance, and median. *Proc. Natl. Acad. Sci. USA* 58(1):66–68.

de la Peña VH, Lai TL, Shao QM (2008) *Self-Normalized Processes: Limit Theory and Statistical Applications*, *Probability and its Applications* (Springer Science & Business Media, Berlin).

Demets DL, Lan KG (1994) Interim analysis: The alpha spending function approach. *Statist. Medicine* 13(13-14):1341–1352.

Even-Dar E, Mannor S, Mansour Y (2002) PAC bounds for multi-armed bandit and Markov decision processes. Kivinen J, Sloan RH, ed. *COLT'02 Proc. 15th Annu. Conf. Comput. Learning Theory* (Springer, Berlin), 255–270.

Fithian W, Wager S (2014) Semiparametric exponential families for heavy-tailed data. *Biometrika* 102(2):486–493.

Foster DP, Stine RA (2008) α-investing: A procedure for sequential control of expected false discoveries. *J. Roy. Statist. Soc. Ser. B Statist. Methodology* 70(2):429–444.

Ghosh BK, Sen PK (1991) *Handbook of Sequential Analysis* (CRC Press, Boca Raton, FL).

Howard SR, Ramdas A, McAuliffe J, Sekhon J (2018) Uniform, nonparametric, non-asymptotic confidence sequences. Preprint, submitted October 18, https://arxiv.org/abs/1810.08240.

James W, Stein C (1961) Estimation with quadratic loss. Neyman J, ed. *Proc. Fourth Berkeley Sympos. Math. Statist. Probab.*, vol. 1 (University of California Press, Berkeley, CA), 361–379.

Jamieson K, Jain L (2018) A bandit approach to multiple testing with false discovery control. Preprint, submitted September 6, https://arxiv.org/abs/1809.02235.

Jamieson K, Malloy M, Nowak R, Bubeck S (2014) lil'UCB: An optimal exploration algorithm for multi-armed bandits. *Proc. 27th Conf. Learning Theory*, *Proc. Machine Learn. Res.*, vol. 35 (PMLR, Barcelona, Spain), 423–439.

Javanmard A, Montanari A (2016) Online rules for control of false discovery rate and false discovery exceedance. Preprint, submitted March 29, https://arxiv.org/abs/1603.09000.

Johari R, Koomen P, Pekelis L, Walsh D (2017) Peeking at A/B tests: Why it matters, and what to do about it. *Proc. 23rd ACM SIGKDD Internat. Conf. Knowledge Discovery Data Mining KDD '17* (Association for Computing Machinery, New York), 1517–1525.

Kalyanakrishnan S, Tewari A, Auer P, Stone P (2012) PAC subset selection in stochastic multi-armed bandits. *Proc. 29th Internat. Conf. Machine Learning* (Omnipress, Madison, WI), 655–662.

Kaufmann E, Cappé O, Garivier A (2014) On the complexity of A/B testing. Preprint, submitted May 13, https://arxiv.org/abs/1405.3224.

Kohavi R, Deng A, Frasca B, Walker T, Xu Y, Pohlmann N (2013) Online controlled experiments at large scale. *Proc. 19th ACM SIGKDD Internat. Conf. Knowledge Discovery Data Mining* (Association for Computing Machinery, New York), 1168–1176.

Lai TL (2001) Sequential analysis: Some classical problems and new challenges. *Statist. Sinica* 11(2):303–351.

Lattimore T, Szepesvári C (2020) *Bandit Algorithms* (Cambridge University Press).

Lehmann EL, Romano JP, Casella G (1986) *Testing Statistical Hypotheses*. Springer Texts in Statistics, vol. 150 (Wiley, New York).

Malek A, Katariya S, Chow Y, Ghavamzadeh M (2017) Sequential multiple hypothesis testing with type I error control. *Proc. 20th*

Internat. Conf. Artificial Intelligence Statist., Proc. Machine Learn. Res., vol. 54 (PMLR, Fort Lauderdale, FL), 1468–1476.

Miller E (2010) How not to run an A/B test. Accessed June 2, 2021, http://www.evanmiller.org/how-not-to-run-an-ab-test.html.

Miller E (2015) Simple sequential A/B testing. Accessed June 2, 2021, http://www.evanmiller.org/sequential-ab-testing.html.

Pollak M, Siegmund D (1975) Approximations to the expected sample size of certain sequential tests. *Ann. Statist.* 3(6):1267–1282.

Robbins H (1970) Statistical methods related to the law of the iterated logarithm. *Ann. Math. Statist.* 41(5):1397–1409.

Robbins H, Siegmund D (1974) The expected sample size of some tests of power one. *Ann. Statist.* 2(3):415–436.

Scott SL (2015) Multi-armed bandit experiments in the online service economy. *Appl. Stochastic Models Bus. Indust.* 31(1):37–45.

Siegmund D (1978) Estimation following sequential tests. *Biometrika* 65(2):341–349.

Siegmund D (1985) *Sequential Analysis: Tests and Confidence Intervals*, *Springer Series in Statistics* (Springer, New York).

Tang D, Agarwal A, O'Brien D, Meyer M (2010) Overlapping experiment infrastructure: More, better, faster experimentation. *Proc. 16th ACM SIGKDD Internat. Conf. Knowledge Discovery Data Mining* (Association for Computing Machinery, New York), 17–26.

Wald A (1945) Sequential tests of statistical hypotheses. *Ann. Math. Statist.* 16(2):117–186.

Yang F, Ramdas A, Jamieson K, Wainwright MJ (2017) A framework for multi-A(rmed)/B(andit) testing with online FDR control. Preprint, submitted June 16, https://arxiv.org/abs/1706.05378.

Zhao S, Zhou E, Sabharwal A, Ermon S (2016) Adaptive concentration inequalities for sequential decision problems. *Adv. Neural Inform. Processing Systems* 29:1343–1351.

Ramesh Johari is a professor in the Department of Management Science and Engineering, and (by courtesy) in the Departments of Computer Science and Electrical Engineering, at Stanford University. His research interests include market design and its application to online platforms, as well as related statistical methods, including experimental design (A/B testing) and online learning.

Pete Koomen co-founded Optimizely in 2010. Before that, he earned a master's in computer science at University of Illinois at Urbana–Champaign.

Leonid Pekelis is head of data science at CloudTrucks, a start-up aimed at digitizing the middle-mile supply chain. He formerly completed his doctorate in statistics at Stanford University with a focus on multiple hypothesis and sequential testing.

David Walsh is a statistician at Unlearn.AI, where he researches methods to improve the efficiency of clinical trials. He completed a PhD in statistics at Stanford University with thesis research on online A/B tests.

Copyright 2022, by INFORMS, all rights reserved. Copyright of Operations Research is the property of INFORMS: Institute for Operations Research and its content may not be copied or emailed to multiple sites or posted to a listserv without the copyright holder's express written permission. However, users may print, download, or email articles for individual use.